TEACHER GUIDE · GRADE 1 · UNPLUGGED SEQUENCING

Plan Your Algorithm

Quick facilitation notes & answer key

Goal

The unit finale: students plan and write a full algorithm for a pattern (a mathematical situation) and for a real task, then improve one by adding a step or fixing the order. Writing algorithms is C3.1; reading their own algorithm and altering it to make it better is C3.2 — both in one task.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil (red/blue crayons optional for the pattern). Optional: shape cards or counters to plan the pattern before drawing.

Run it (about 20-25 min)

1. I DO: model planning a pattern out loud — 'circle, square, square… I'll repeat it: circle, square, square.' Write the 6 steps, then draw.

2. WE DO: Exercise 1 — students plan their own 6-shape pattern, write the steps, and draw it.

3. YOU DO: Exercise 2 — write a real job algorithm (4-6 steps in order). Exercise 3 — improve it by adding a step or fixing the order, and explain why.

4. Celebrate: 'You can read, write, follow, plan, and improve algorithms — you are a coder!'

Answer key (modelled by the run-gate)

Ex 1 — open; a sample is an ABB pattern: ● ■ ■ ● ■ ■ (steps: box1 ●, box2 ■, box3 ■, box4 ●, box5 ■, box6 ■). Accept any pattern where the steps match the drawing.

Ex 2 — open; a sample 'pack your backpack': open the backpack, put in your books, put in your lunch, zip it up, put it on. Accept any 4-6 sensibly ordered steps.

Ex 3 — open; a sample improvement: add 'check your homework is in' before 'zip it up', because once it's zipped you'd have to undo it. Accept any added step or reorder with a reason.

Watch for & differentiate

Common slip in Ex 1: the written steps don't match the drawing — have students read their steps aloud while pointing at each box.

Simplify: give a 4-shape pattern and a 3-step job for students who need a smaller load.

Extend: ask students to swap finished algorithms and follow a partner's steps to test that they work.

Success indicator: the child writes a pattern algorithm that matches their drawing, a correctly-ordered job algorithm, and a sensible improvement with a reason.